

TECHNICAL SPECIFICATION & ARCHITECTURE MANUAL

OCEAN PRO

Real-Time Interactive Marine Ecosystem, Bathymetric Ocean Simulation,
Oceanographic Instruments & Multi-Agent Artificial Intelligence System

ARCHITECTURE

WebGL / Three.js + Express + Google Gemini API

SCIENTIFIC GROUNDING

WoRMS, FishBase, OBIS, NOAA ERDDAP

SIMULATION LAYER

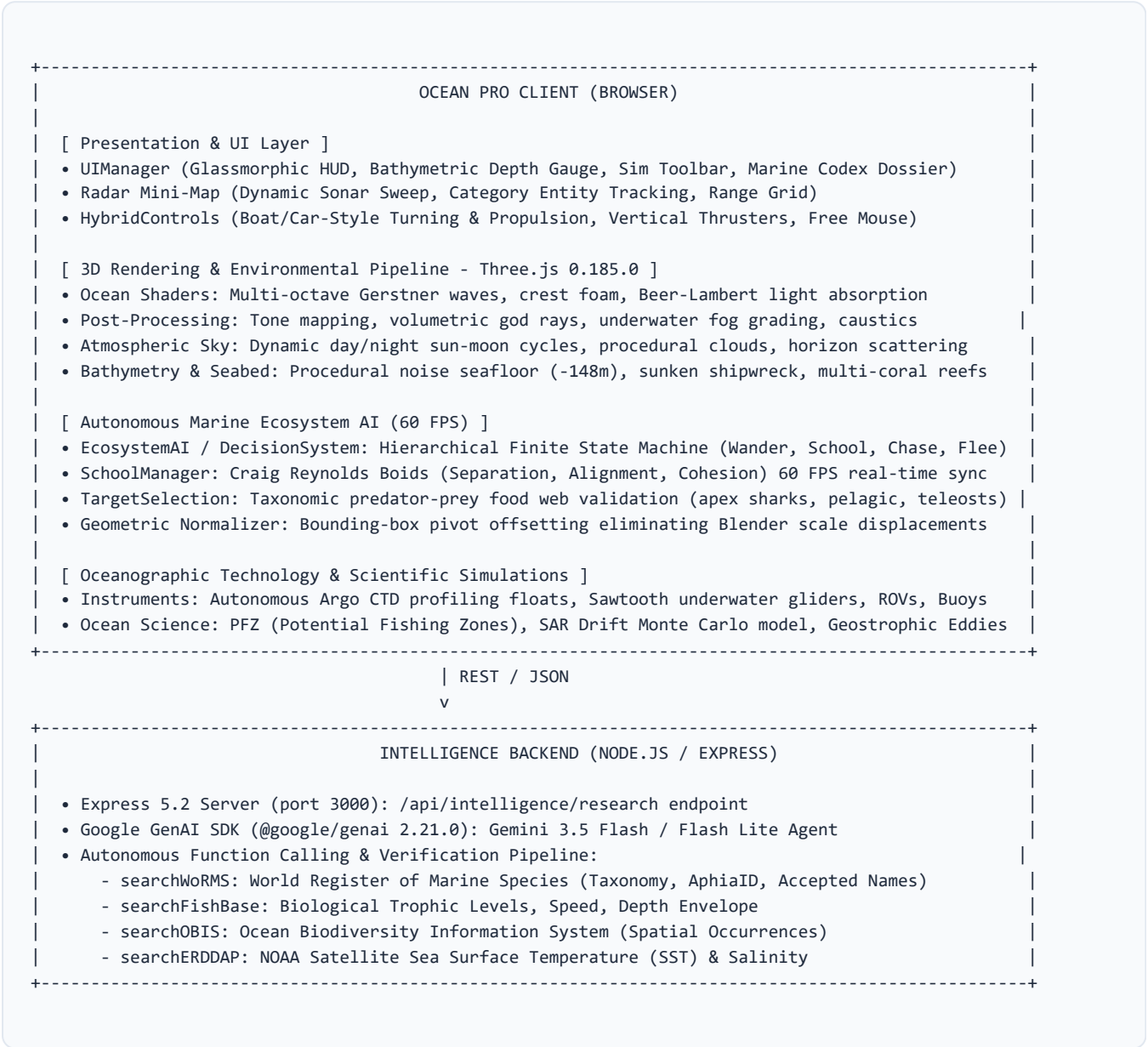
FFT/Gerstner Waves, Boids Schooling, HFSM AI

VERSION & RELEASE

Version 1.0.0 (Production Architecture)

1. Executive Architecture Overview

Ocean Pro is an end-to-end, scientifically grounded 3D oceanographic simulation and autonomous marine ecosystem. It pairs high-performance WebGL graphics (physical water shaders, caustic projection, light attenuation, and volumetric atmosphere) with high-fidelity agent artificial intelligence and authentic global marine science data.



Key Engineering Paradigm

Ocean Pro separates **instant local execution** from **background scientific enrichment**. The client maintains an authoritative local Marine Knowledge Base (`MarineKnowledgeBase.js`) providing zero-latency biological profiles from frame 0, while asynchronous background research via the Gemini backend enriches and validates species metadata via live WoRMS and FishBase queries without hitching the 60 FPS WebGL render loop.

2. Complete Technology Stack

The table below summarizes every core library, runtime, framework, and tooling used in Ocean Pro:

Technology	Version	Category	Role in Application
Three.js	<code>^0.185.0</code>	3D Engine	WebGL core renderer, custom GLSL shader materials, scene graph, lights, cameras, SkinnedMesh skeletal animation.
@google/genai	<code>^2.21.0</code>	AI / LLM SDK	Official Google GenAI SDK interfacing with Gemini 3.5 Flash for autonomous tool calling, species taxonomy research, and evidence synthesis.
Express	<code>^5.2.1</code>	Backend Runtime	REST API server proxying client requests, orchestrating multi-turn tool calling with Gemini, and protecting API keys.
Vite	<code>^6.0.0</code>	Build Tool	Development server with Hot Module Replacement (HMR) and optimized Rollup-based production bundler with GLTF/GLB asset hashing.
lil-gui	<code>^0.20.0</code>	HUD / GUI	Interactive runtime developer controls for wave parameters, time-of-day sunlight angles, underwater turbidity, and debug states.
cors	<code>^2.8.6</code>	Middleware	Enables secure cross-origin resource sharing between Vite frontend (port 5173) and Express backend (port 3000).
dotenv	<code>^17.4.2</code>	Environment	Zero-dependency module that loads environment variables (<code>GEMINI_API_KEY</code>) from <code>.env</code> files into <code>process.env</code> .
concurrently	<code>^10.0.5</code>	Dev Tooling	Runs Vite client server and Node Express intelligence server simultaneously under a single <code>npm run dev</code> lifecycle.
HTML5 / Canvas	Standard	Graphics	High-performance 2D Canvas rendering for the bathymetric Sonar Radar HUD, sweep lines, and blip rendering.
CSS3 (Modern)	Standard	UI Styling	Backdrop-filter blur glassmorphism, responsive grid layouts, custom scrollbars, and keyframe HUD animations.

3. Feature Breakdown: "For What Feature We Used What"

3.1. Photorealistic Ocean Water & Underwater Optical Atmosphere

Feature Component	Technologies & Modules Used	Implementation Details
Wave Synthesis	Custom GLSL Vertex Shader, Three.js ShaderMaterial	Multi-octave Gerstner wave equations running on the GPU. Computes analytical wave steepness, wavelength, direction, and displacement vector with normal Jacobian calculation.
Water Surface Optics	Custom GLSL Fragment Shader, Three.js PlaneGeometry	Fresnel reflection, Cook-Torrance specular highlights, dynamic foam generation at crests based on Jacobian compression thresholds, and normal map perturbation.
Underwater Caustics	Three.js TextureLoader , Additive Blending, GLSL	Dynamic high-contrast caustic light maps projected downward onto the seabed and marine life bodies, fading with depth according to exponential light absorption.
Light Absorption & Turbidity	Beer-Lambert Law formulation, Three.js FogExp2	Wavelength-dependent absorption: red light attenuates within 5m, yellow/green within 25m, leaving deep cyan/blue hues down to the aphotic seabed at -148m.
Volumetric God Rays	Three.js Post-Processing, Custom Pass in Post.js	Screen-space radial blur pass taking bright sun specular samples and projecting luminous volumetric shafts through the water column.

3.2. Camera Navigation & Vehicular Steering

Feature Component	Technologies & Modules Used	Implementation Details
Boat / Car-Style Steering	HybridControls.js , Vector Math, Euler Angles	A/D Keys: Perform continuous heading rotation without lateral strafing. W/S Keys: Propel along forward/backward vector. W+A / W+D: Produce curved arcs.
Vertical Propulsion	HybridControls.js , Vertical Kinematics	Q/E and Shift/Space allow smooth diving under the thermocline, exploring deep abyssal reefs, or resurfacing with water-transition detection.
Pointer Freedom	DOM Mouse Event Listeners, Three.js Raycaster	Pointer lock is intentionally avoided so the cursor is always available to click Codex cards, radar, advisory filters, and toolbar controls. Mouse drag orbits view.

3.3. Autonomous Marine Life & Multi-Agent Ecosystem AI

Feature Component	Technologies & Modules Used	Implementation Details
Model Ingestion & Centering	Three.js <code>GLTFLoader</code> , <code>SkeletonUtils</code>	Surgically offsets model bounding-box centers (<code>pivotGroup</code>) to eliminate Blender coordinate offsets, guaranteeing all creatures stay sub-surface ($Y \leq -2.5\text{m}$).
Skeletal Animation	Three.js <code>AnimationMixer</code> , Clip matching	Detects swim clips from GLTF animations (<code>findSwimClip</code>), applying speed-synchronized cadence playback (<code>timeScale</code>) so tail frequencies match propulsion.
Real-Time Schooling	<code>SchoolManager.js</code> , <code>Steering.js</code>	Craig Reynolds Boids algorithm executing every frame at 60 FPS. Groups yellowback fusiliers, tangs, and discus with continuous flocking synchronization.
Predator-Prey Food Web	<code>TargetSelection.js</code> , <code>DecisionSystem.js</code>	Evaluates prey eligibility against predator taxonomy (<code>huntsTaxa</code>). Triggers predator <code>CHASE</code> and <code>ATTACK</code> lunges, while prey trigger evasive <code>FLEE</code> zig-zag bursts.
Depth & Habitat Clamping	<code>OceanEnvironment.js</code> , <code>EcosystemAI.js</code>	Strict biological depth clamping prevents creatures from breaching above -2.5m or penetrating through the seabed floor at -148m .

4. External APIs & Web Standards Used

Ocean Pro leverages both industry-standard browser APIs and real-world scientific marine data APIs:

API / Service	Endpoint / Integration	Protocol	Purpose & Data Retrieved
Google Gemini API	<code>GoogleGenAI</code> SDK (Interactions / Chat API)	HTTPS / gRPC via SDK	Agentic AI reasoning, 3D asset filename parsing, autonomous tool orchestration, and biological evidence synthesis.
WoRMS REST API	<code>marinespecies.org/rest/AphiaRecordsByMatchNames</code>	HTTPS GET REST	World Register of Marine Species. Authoritative taxonomic hierarchy (Kingdom, Phylum, Class, Order, Family, Genus), accepted scientific names, and AphiaIDs.
FishBase API	<code>fishbase.ropensci.org/species</code>	HTTPS GET REST	Comprehensive marine biology database: trophic level (herbivore/carnivore), biological swimming speed, depth envelopes, and IUCN conservation status.
OBIS API	<code>api.obis.org/v3/occurrence</code>	HTTPS GET REST	Ocean Biodiversity Information System. Global geographic occurrence records, lat/long density centroids, and ocean basin habitats.
NOAA ERDDAP API	<code>coastwatch.pfeg.noaa.gov/erddap/</code>	HTTPS GET REST	NOAA satellite SST (Sea Surface Temperature), ocean salinity, and geostrophic surface currents for real-world environmental conditioning.
WebGL 2.0 API	Browser standard via Three.js	Native WebGL	GPU rendering context, floating-point framebuffers, instanced mesh rendering for reefs, uniform buffer objects, and GLSL ES 3.0 shaders.
HTML5 Canvas 2D API	<code>HTMLCanvasElement.getContext('2d')</code>	Native DOM	Renders the circular bathymetric Sonar Radar HUD, sweep beam rotation, blip coordinate mapping, and vessel heading indicator.
Web Storage API	<code>window.localStorage</code>	Client Storage	Persistent client-side caching of verified species profiles in <code>KnowledgeCache.js</code> to ensure instant load times and eliminate API rate limits.

5. Oceanographic Instrumentation & Scientific Simulations

5.1. Autonomous Instruments

- **Argo Profiling Floats (`ArgoManager.js`)**: Modeled after the International Argo Program. Floats drift neutrally at 1,000m, execute deep vertical profiling casts down to 2,000m, and ascend to the surface to transmit CTD (Conductivity, Temperature, Depth) telemetry.
- **Autonomous Underwater Gliders (`GliderManager.js`)**: Modeled after NOAA IOOS ocean gliders. Utilize internal mineral oil bladders for buoyancy-driven sawtooth flight profiles through coastal and pelagic transects without propellers.
- **Benthic Shipwrecks & Habitats (`ShipwreckManager.js`)**: Sunken galleon asset positioned on the abyssal seafloor (-146m) acting as an artificial reef congregating demersal species.

5.2. Scientific Simulation Modules

- **Potential Fishing Zone (PFZ) Advisory (`PFZEngine.js`)**: Simulates oceanic thermal gradients, SST front delineation, and chlorophyll concentration boundaries where pelagic fish congregate.
- **SAR (Search and Rescue) Leeway Drift Engine (`DriftSimulationEngine.js`)**: Monte Carlo drift dispersion calculation integrating surface current vectors, 10-meter windage vectors, and Stokes wave drift to project datum search areas.
- **Geostrophic Eddies (`EddySimulation.js`)**: Simulates cyclonic (cold core, upwelling) and anticyclonic (warm core, downwelling) ocean vortices.

6. Codebase File Structure

```
d:/ocean-pro/
├─ package.json           # Project dependencies and npm scripts
├─ vite.config.js         # Vite build and asset pipeline configuration
├─ index.html             # Main entry HTML with viewport and DOM mounts
├─ server/
│   └─ index.js           # Express 5 backend server (Port 3000)
│   └─ intelligence/
│       └─ GeminiAgent.js  # Server-side Gemini AI research agent
│       └─ tools/
│           └─ index.js    # Tool exports and registry
│           └─ worms.js    # WoRMS REST API integration
│           └─ fishbase.js # FishBase biological API integration
│           └─ obis.js     # OBIS biogeographic API integration
│           └─ erddap.js   # NOAA ERDDAP oceanographic data API
└─ src/
    └─ main.js            # Core app entrypoint and main animation loop
    └─ Ocean.js           # Three.js ocean water mesh and Gerstner shaders
    └─ Sky.js             # Atmospheric dome, sun, and moon positioning
    └─ Clouds.js          # Procedural volumetric clouds
    └─ Floor.js           # Noise-displaced seabed terrain
    └─ Post.js            # Custom post-processing and volumetric lighting
    └─ Radar.js           # Canvas 2D bathymetric sonar radar HUD
    └─ HybridControls.js  # Boat/car-style camera navigation system
    └─ creatures/
        └─ CreatureLoader.js # GLTF loader, pivot centering, and scaling
        └─ CreatureRegistry.js # Spatial entity registry
    └─ ecosystem/
        └─ EcosystemAI.js   # Agent lifecycle coordinator (60 FPS)
        └─ DecisionSystem.js # Hierarchical FSM (Wander, School, Chase, Flee)
        └─ SchoolManager.js # Boids flocking coordination algorithm
        └─ TargetSelection.js # Food web predator-prey taxonomy matching
        └─ Steering.js      # Kinematic steering behaviors
        └─ Perception.js    # Entity sensory radius and visibility
        └─ CreatureMemory.js # Agent state, targets, and cooldown timers
        └─ BehaviorBuilder.js # Profile constructor from biological knowledge
        └─ BehaviorProfile.js # Formalized creature behavioral profile model
    └─ environment/
        └─ CoralManager.js  # Benthic reef distribution and seabed anchoring
        └─ ShipwreckManager.js # Seafloor galleon placement and reef habitat
        └─ OceanEnvironment.js # Physical bounds and depth clamping
    └─ instruments/
        └─ InstrumentManager.js # Coordinator for robotic ocean instruments
        └─ ArgoManager.js      # Autonomous CTD profiling float behavior
        └─ GliderManager.js    # Autonomous underwater glider sawtooth flight
    └─ intelligence/
        └─ KnowledgeBuilder.js # Multi-tier knowledge orchestration layer
        └─ MarineKnowledgeBase.js # Authoritative 40+ species local encyclopedia
        └─ KnowledgeCache.js   # LocalStorage profile persistence
        └─ GeminiAgent.js      # Client proxy to backend intelligence
    └─ simulation/
        └─ PFZEngine.js       # Potential Fishing Zone simulation
        └─ DriftSimulationEngine.js # SAR Leeway Monte Carlo drift engine
    └─ ui/
```

```
└─ UIManager.js      # Glassmorphism UI, Codex modal, and depth HUD
```